

INFO-I300 HCI/d

Project 2

AAV (Automatic Adjusting Volume)

Auto-Adjusting Volume Technology through Ambient Noise Sensors

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Rationale and Research

Like everyone who enjoys a nice sit down at a cafe on a Sunday afternoon getting ready to study for their classes the following week, the first thing we do is to close out the external noise by turning on our favorite playlist. I personally enjoy listening to music while studying, but there’s always a small problem with music playing in the background.

We’ve all encountered this moment, you’re in a train of thought, all of a sudden, a catchy lyric or tune whizzes your mind off of that brilliant epiphany and gets you humming. Next thing you know, that all-important spark has been pulverized into smithereens and you’re back at ground zero.

This notion of music being a distraction is exemplified in an article by the Guardian, where it claims that lyrics take up processing space, which could interfere with the material at hand (Source 16). Yet despite the argument of music being a distraction, people still persist on studying or working with music on. From the four people I interviewed about questions regarding studying with music, all four of the people responded saying that they either studied with music on at all times or sometimes depending on how focused they wanted to be (Source 1, 2, 3, 4).

There are contrarian beliefs that music can be helpful while studying. An article from Wired about the study of neurosciences of music indicates that music helps us release dopamine, which promotes happiness (Source 11). Another article from USA today takes a step further and claims that music puts us in an induced state to help reduce stress and lead to a therapeutic effect leading to a upsurge in cognitive performance (Source 14).

This made me wonder, how could there be a way of having music on, but not make it distractive. As I pondered upon an article from the Harvard Business Review, where it claimed that “the right level of ambient noise triggers our minds to think more creatively”, this idea that music could coexist within a study environment triggered me (Source 17).

How Can We Solve this Issue?

Some might argue that the problem can be alleviated with noise cancelling headphones, but a decent option (like the one shown in Image 6) could climb above the \$300 threshold. Truth is not all individuals want to be isolated from the noise and are willing to pay that amount for a pair of headphones (Source 9, 33). A study conducted from the Journal of Consumer Research suggested that a background noise of roughly 70 decibels could strike the balance and possibly encourage people to think more creatively and work efficiently (Sources 12, 21, 23).



Image 6: Wireless Noise Cancelling Headphones



Image 7: Ambient Noise Sensor

| Noise                                                                     | Average decibels (dB)                  |
|---------------------------------------------------------------------------|----------------------------------------|
| Leaves rustling, soft music, whisper                                      | 30                                     |
| Average home noise                                                        | 40                                     |
| Normal conversation, background music                                     | 60                                     |
| Office noise, inside car at 60 mph                                        | 70                                     |
| Vacuum cleaner, average radio                                             | 75                                     |
| Heavy traffic, window air conditioner, noisy restaurant, power lawn mower | 80–89 (sounds above 85 dB are harmful) |
| Subway, shouted conversation                                              | 90–95                                  |
| Boom box, ATV, motorcycle                                                 | 96–100                                 |

Image 9: Chart of Decibel Noise Levels

## Today's Solutions and Trends

The trend today is geared towards wireless technology. In the fourth quarter of 2018 alone, wireless hearables reached a figure of over 12.5 million units sold globally (Source 31). The flexibility in both function, design, and the longevity of battery life has made opting for wireless peripherals a popular choice for consumers (Source 32). Features such as motion detectors and noise cancelling have also made earphones a far more efficient and less cumbersome device to use (Source 9, 19, 25).

Currently there are multiple ways of handling the issue of music becoming a distraction while studying, but there are lingering questions that should be taken into consideration when taking solutions into account:

- Adjust volume to a minimum: How are we certain that you get the right volume, changing it constantly would affect the momentum of studying.
- Noise cancelling headphones: Pricey, average prices for a highly rated active noise cancelling headphones could cost over \$300 (Source 9, 33).
- Turn the music off: What about external noises, what if it gets too loud and interfere with the work at hand?
- Listen to classical music: What if the person doesn't enjoy classical music?
- Listen to music from a different language: Why turn on new music with unfamiliar lyrics that could cause more distraction?

### Proposition: Auto-Adjusting Volume Technology through Ambient Noise Sensors

I believe that the ANS (Ambient Noise Sensing) sensors (shown in Image 7) have substantial potential and applications. The idea is to install ambient noise sensors to the microphones of the earphones/headphones which then identifies the noise levels surrounding the listener (through decibels), which is then processed by the onboard computer of the earpiece to auto adjust the volume through software and cater to the listener's preference. This feature (the AAV technology) can be toggled by the user if they want the earphone to auto adjust the volume depending on the environment.

Why ambient noise sensors? The point is for the listener to seamlessly focus on their work, without having to adjust on the go. The ambient noise sensors will actively detect the surrounding noise levels around the area and immediately tune the volume to the optimum sound that the user wants it to (Source 24, 25). Every listener has their optimum volume, which this technology can be adjusted to, meaning the user sets the benchmark sound level as the one that they want the technology to pivot around.

## Technologies Utilized

| Ambient Noise Detecting Sensors                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Volume Adjusting Algorithm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• This sensor will be attached to work in tandem with the microphone (which will be attached at a location that is optimum for reading external noise).</li><li>• Main component of this technology is used as a way to detect and monitor ambient noise that is occurring around the sensor (in this case the earphone when it is being utilized by the user). The sensor detects the noise (decibel levels), which is then processed by the algorithm to adjust the volume to counteract the ongoing external noise (Source 25, 27). This occurs perpetually so that the user doesn't have to adjust the volume themselves.</li></ul> | <ul style="list-style-type: none"><li>• The algorithm will automatically adjust the volume according to the environment based on the decibel levels that the sensor reads out. The volume won't be constantly changing (since sound levels can vary by the second) but will change depending on the disparity of the noise levels at certain periods of time and location. The "prime" volume can be toggled by the user's preference since sound is subjective.</li><li>• Though the sensor can be utilized as a way to cancel out noise, this technology (in this case) will solely be a supplement to the software feature that will help the automatically adjust the volume (Source 25).</li></ul> |

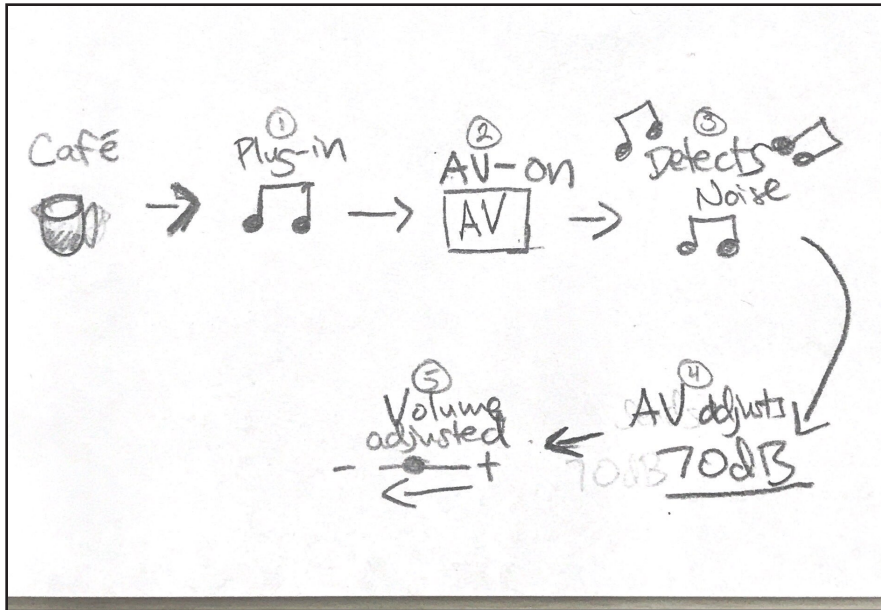


Image 3: Storyboard of Technology in Action

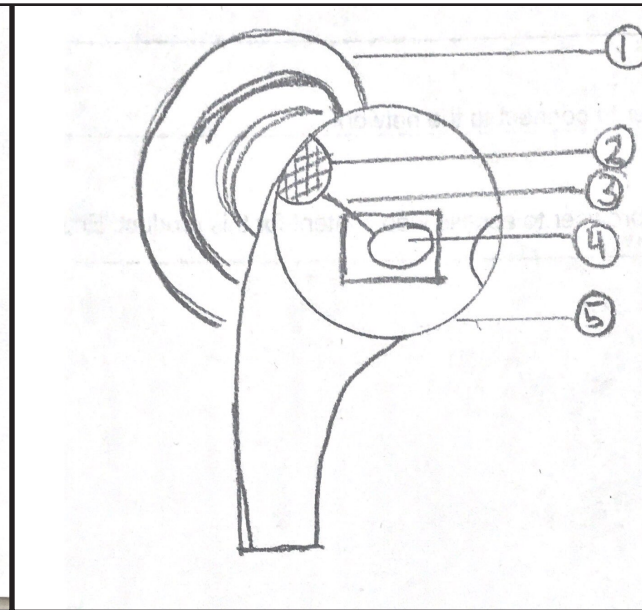


Image 1: AAV Implemented in an Airpod

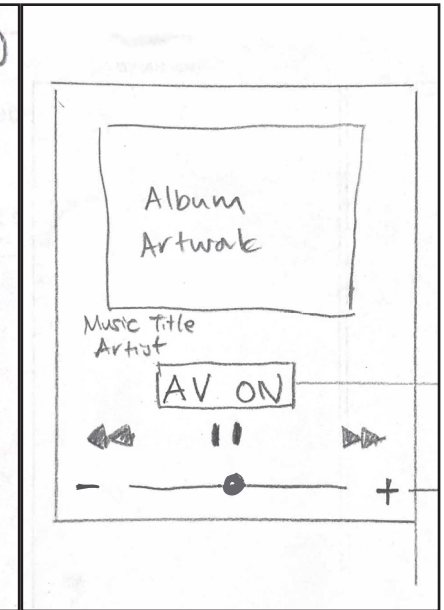


Image 4: UI of Music Player with AAV Technology Implemented

Image 1 illustrates the mockup of when the ANS sensor is applied to the earpiece (in this case an Apple Airpod):

1. The circulation illustrates the sound that is being monitored and filtered by the sensor (this telemetry is then communicated with the onboard computer inside the earpiece).
2. The ambient noise sensor, which is built into the microphone takes in the sound and monitors the decibel levels.
3. The wire connecting the sensor with the onboard computer.
4. The onboard computer, which processes the readings of the sound sensor, automatically readjusts the volume when the toggle (AAV) is turned on.
5. The earpiece, which houses all of the features inside the earphone (in this case the technology is placed into an Apple Airpod).

The UI (Image 4) illustrates the simplicity in regard to how the AV would function, a toggle can be pressed to activate and deactivate, leaving it up to the user to decide whether he/she wants to utilize it. The wireframe (Image 2) illustrates a constant interaction of both the earphone and the phone working in conjunction with one another. It's a reciprocation, not a one-way streak, the technology is constantly listening to the overall environment to readjust and align with what the right volume should be at the given period.

The Storyboard (Image 3) illustrates the scenario of how the technology would be put into action in an actual live situation:

1. Before everything happens, the user goes to a destination, in this case a cafe. He/she then plugs in her wireless earphone.
2. The Automatic Adjusting Volume function is toggled on and the sensor starts to activate.
3. The sensor detects the ambient noises in the background and triggers the AAV (automatic adjusting volume) technology.
4. After the sensor reads the ambient noise, the technology adjusts the volume according to the external decibel (sound) level.
5. The algorithm automatically takes into account the ambient noise levels and adjusts the volume to match the optimum sound level that the user has identified. Since volume levels are subjective, the feature (AAV) can be readjusted to the user's preference.



Photo 1: The Pourhouse Cafe 1



Photo 2: The Pourhouse Cafe 2

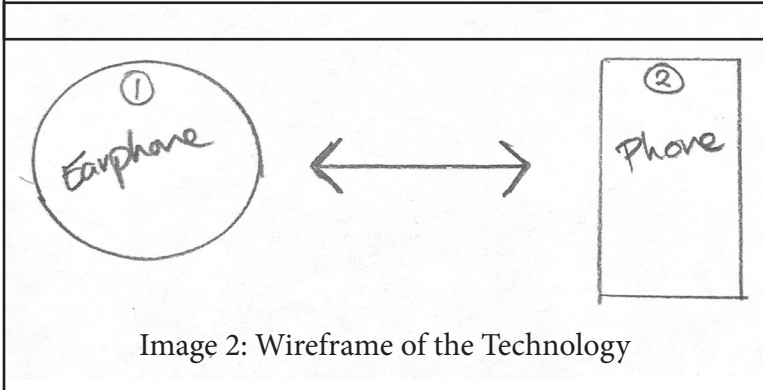


Image 2: Wireframe of the Technology

## Findings and Takeaways from Surveys and Observations

**Background on the interviews:** The nine questions plus the extra questions were designed to scrutinize whether individuals not only preferred to listen to music, but what types of music they listened to while they studied. As the questions progressed, I added open-ended questions to determine specific features that they wished they had in their current earphone/headphones (Source 1, 2, 3, 4).

**Background on Observations:** The observation gave me the opportunity to see how people utilized their earphones in a study setting. Throughout the observation, I was able to analyze and dissect the mannerisms and behavior of people when they were interacting and utilizing their earphones/headphones (Source 5, 6, 20).

**Identifying Patterns:** The overarching pattern of the results were that the respondents listen to music while studying if it won't interfere with their work. The interviews indicate that listening to familiar music helps individuals focus, because they don't have to think about the music. During my time observing, I noticed majority of the individuals that had hearables on were glued to their screen and sitting at a hunched position to the point that their peripheral vision was at a bare minimum. This gave me the idea that social interaction was kept at a minimum and the focus was solely down to the task at hand, studying.

## The Limitations of the Interviews and Observations

Both interviews and observations suffered from the lack of a larger sample size to provide a dataset that would allow a more holistic and bigger scope to tackle the issue. Interviewing college students and observing a cafe gives a very limited outlook on the effect hearables have on individuals while studying (Source 1, 2, 3, 4, 5, 6).

## What's Next and Future Applications

Despite the applications of the AAV technology there are still areas and questions to be asked and considered in regard to feasibility and substitutions that could be used:

- How much battery life is the ANS (ambient noise sensor) going to sap away and how will this affect performance in the long run?
- Will the AAV affect how the product is designed; how small is the sensor?
- Can this feature be entirely through software instead of having to add another sensor? Since microphones offer a similar role, can a software patch be layered on top where decibel levels are processed through another app (such as a decibel app).
- What is the range and boundaries of the ANS? In order to make the sensors be more efficient, how far are we going to limit the sensors in terms of volume of the location (certain places have higher ceilings than others).
- Sensors can detect different sorts of external noise such as natural and external projection noises, how will the sensor identify and differentiate them?

## Works Cited

### Primary:

Surveys and Observations are Located in Appendix 1-6:

#### Survey Questions:

1. Do you listen to music while studying? (Y/N)
2. What type of music do you listen to while studying (If the answer is no, go to 2a)?
- 2a. Why don't you listen to music while studying?
3. When you listen to music while studying, does it help you (If the answer is no, go to 3a)?
- 3a. Why do you think not listening is better for you while you study?
4. How loud do you listen to your music while studying?
5. Why do you listen to music while studying?
6. What feature do you wish your earphones had?
7. Do you have a preset playlist while studying?
8. Do you listen to songs that you're familiar with or are you more adventurous with your choice of music?
9. How often do you listen to music?

#### Extra Questions (yes or no questions):

1. Would a quieter environment help you concentrate?
2. Would you mind if your earphones auto adjusted the volume for you?
3. When you study do you stay at one place for a long period of time?
4. Would you say music helps you concentrate on your work?
5. When you're looking for a place to study, do you like to isolate yourself?

Source 1: Interview of Chaewon Lee conducted by Chae Woon Kim on 10/11/2019 (Appendix 3)

Source 2: Interview of Elise Jane Lee conducted by Chae Woon Kim on 10/14/2019 (Appendix 4)

Source 3: Interview of Heemean Kim conducted by Chae Woon Kim on 10/13/2019 (Appendix 5)

Source 4: Interview of Megan Jerrells conducted by Chae Woon Kim on 10/13/2019 (Appendix 6)

Source 5: Observation of Earphones/Headphones Interaction in a Cafe conducted by Chae Woon Kim on 10/12/2019 (Appendix 1)

Source 6: Observation of Cafe Decibel Readings using the Decibel X App conducted by Chae Woon Kim on 10/19/2019 (Appendix 2)

### Images:

Image 1: AAV (Automatic Adjusted Volume) Technology Implemented in an Apple AirPods (source: Chae Woon Kim 10/19/2019)

Image 2: Wireframe of AAV Technology in Action (source: Chae Woon Kim 10/19/2019)

Image 3: Storyboard of AAV Technology in Action (source: Chae Woon Kim 10/19/2019)

Image 4: UI of Music Player with AAV Technology Implemented (source: Chae Woon Kim 10/19/2019)

### Photos:

Photo 1: The Pourhouse Café 1 (source: Chae Woon Kim 10/12/2019)

Photo 2: The Pourhouse Café 2 (source: Chae Woon Kim 10/12/2019)

## Works Cited Continued

### Secondary:

#### Sources:

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Sources 9: Are Noise Cancelling Headphones Worth It? (<https://www.forbes.com/sites/geoffreymorrison/2014/02/21/are-noise-cancelling-headphones-worth-it/#228606c246d5>, 10/20/2019)

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Sources 13: Benefits of Studying with Music (<https://www.fnu.edu/benefits-studying-music/>, 10/20/2019)

Sources 14: Health Benefits of Music (<https://www.usatoday.com/story/news/health/2013/12/17/health-benefits-music/4053401/>, 10/20/2019)

Sources 15: Music Moves Brain to Pay Attention (<https://med.stanford.edu/news/all-news/2007/07/music-moves-brain-to-pay-attention-stanford-study-finds.html>, 10/20/2019)

Sources 16: Listening to Music Hinders Learning (<https://www.theguardian.com/teacher-network/2018/mar/14/sound-how-listening-music-hinders-learning-lessons-research>, 10/19/2019)

Sources 17: Office Space Versus Open Office (<https://hbr.org/2017/10/why-you-can-focus-in-a-coffee-shop-but-not-in-your-open-office>, 10/20/2019)

Sources 18: Apple AirPods (<https://www.apple.com/airpods/>, 10/20/2019)

Sources 19: Apple AirPods Features ([https://support.apple.com/kb/SP792?viewlocale=en\\_US&locale=en\\_US](https://support.apple.com/kb/SP792?viewlocale=en_US&locale=en_US), 10/20/2019)

Sources 20: Crowded Coffee Shops Fire Up Creativity (<https://www.theatlantic.com/health/archive/2012/06/study-of-the-day-why-crowded-coffee-shops-fire-up-your-creativity/258742/>, 10/20/2019)

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Sources 23: Impact of Decibel Levels (<https://myhealth.alberta.ca/Health/Pages/conditions.aspx?hwid=tf4173>, 10/17/2019)

Sources 24: Ambient Noise Sensor Assessment Report (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5426841/>, 10/17/2019)

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Sources 27: Ambient Noise Sensor Product Manual (<https://www.bogen.com/products/pdfs/specialelectronicspdfs/ANS501m.pdf>, 10/17/2019)

Sources 28: Trend of Coffee Shops (<https://phys.org/news/2017-08-trend-coffee.html>, 10/17/2019)

Sources 29: Popularity of AirPods (<https://www.cbsnews.com/news/apple-airpods-are-so-beloved-people-spend-more-than-half-a-billion-replacing-them/>, 10/20/2019)

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Source 31: Popularity in Wireless Hearables (<https://www.counterpointresearch.com/global-true-wireless-hearables-market-reaches-12-5-million-units-q4-2018/>, 10/20/2019)

Source 32: Why Wireless? (<https://www.theverge.com/2018/8/31/17803728/wireless-headphones-usb-c-google-assistant-siri-ifa-2018>, 10/20/2019)

Source 33: Top Noise Cancelling Headphones (<https://www.cnet.com/news/best-noise-cancelling-headphones-of-2019/>, 10/20/2019)

#### Images:

Image 5: Apple AirPods Wireframe from September 2016 Apple Keynote ([https://youtu.be/NS0txu\\_Kzl8?t=5305](https://youtu.be/NS0txu_Kzl8?t=5305) (Time: 1:28:24), 10/18/2019)

Image 6: Wireless Noise Cancelling Headphones ([https://www.bose.com/en\\_us/products/headphones/over\\_ear\\_headphones/quietcomfort-35-wireless-ii.html?mc=25\\_PS\\_HP\\_PL\\_00\\_GO\\_&gclid=EAIaIQobChMIv5up0uyr5QIVBNvACh0I4wCKEAQYABEGKQiPD\\_BwE&gclsrc=aw.ds#v=qc35\\_ii\\_silver](https://www.bose.com/en_us/products/headphones/over_ear_headphones/quietcomfort-35-wireless-ii.html?mc=25_PS_HP_PL_00_GO_&gclid=EAIaIQobChMIv5up0uyr5QIVBNvACh0I4wCKEAQYABEGKQiPD_BwE&gclsrc=aw.ds#v=qc35_ii_silver), 10/17/2019)

Image 7: Ambient Noise Sensor (<https://www.asl-control.co.uk/product/ambient-noise-sensor/>, 10/20/2019)

Image 8: Wireless Earphones (<https://www.runnersworld.com/gear/a20852590/the-best-wireless-headphones-for-running/>, 10/20/2019)

Image 9: Chart of Decibel Noise Levels (<https://myhealth.alberta.ca/Health/Pages/conditions.aspx?hwid=tf4173>, 10/20/2019)

## Appendix 1

### Observation 1 (Source 5)



Photo 1: The Pourhouse Cafe 1



Photo 2: The Pourhouse Cafe 2

The Pourhouse 1:00 - 4:00 pm, Café, Sunday (10/12/2019)

The purpose of this observation was to find out the way how people interacted and utilized their earphones/headphones. This analysis was to also observe a cafe at peak hour and determine the difference in behavior when people were either using or not using their earphones/headphones.

The Timeframe: 1:00 – 4:00pm, on Sunday (10/12/2019)

The Environment: Dimly lit, but enough light for individuals to study.

Questions that I'm asking myself as I am conducting the observation:

- When people are using their earphones, how are they reacting to the environment?
- How are people interacting with their environment?
- What is their body posture when they put their earphones on?
- How glued are they to their work?
- Are people using their earphones/headphones and for what reason?

Total People within the Vicinity at Current Time (2:30pm): 18

Classifying Individuals and How They Behaved:

Conversing (2): Not listening to music and conversing with other people, fluid body language, facial expressions are varying by the second.

Flexible (5): Not too fixated on their screens, either watching a video or looking around and conversing with others. Earphones are taken on and off during various periods. Short term (3-5 minute) interactions with people around them.

Glued (11): Solely focused on their screen. Music was solely for study purposes (if they were wearing earphones) (as there weren't any leaking noises, which indicates a low enough volume) to cover out distractions and external noise. Hunched posture and wasn't fazed by anything occurring around

## Appendix 1, 2

### Observation 1 Continued (Source 5)

Out of a possible 18 people, 16 people (the one's that were "flexible" or "glued") were wearing their earphones for the duration of the three hours I was conducting the observation. All of them preferred wireless over wired earphones. No one wore headphones.

The two ("Conversing") that were conversing were on the same table talking to one another, sitting at a straight 90-degree posture, both their hands were waving and moving around in perpetual motion as they were conversing with one another.

The people that were "flexible" sat amongst others and found time to speak to the people around them. Most of the time they were either watching a video or interacting with one or more devices such as a laptop or a phone.

People that were "glued" were focused on their own work and for the majority of the time deviated from social interaction. The only time they interacted was when the phone rang and when they were going to the bathroom. The rest of the time for the "glued" individuals were spent either focusing on their work, or taking a sip of their coffee; body posture was at a hunched position to the point that the individual's peripheral vision was solely the screen and their notebooks.

From the observation, it is apparent that when an individual has headphones or earphones are on, they're mostly glued to their screens or the task that they're focusing on.

Observation 2: This observation was to determine the sound levels of the Soma Café (the Pourhouse was closed), conducted on 10/19/2019. I chose a similar remote café that is both quiet (a place to study) and dimly lit, to stay as consistent with the first observation as much as possible. This sound analysis was used as an indicator of noise levels to supplement the first observation. On average the sound was roughly 60 dB from a span of a minute, but during my time observing, it varied from a high of 91 dB to a minimum of 50 dB (Source 30).

### Observation 2 (Source 6, 30)



#### Measurement Report

**Time:** 2019/10/19 5:11 PM

**Device:** iPhone 7 Plus, iOS 13.1.2

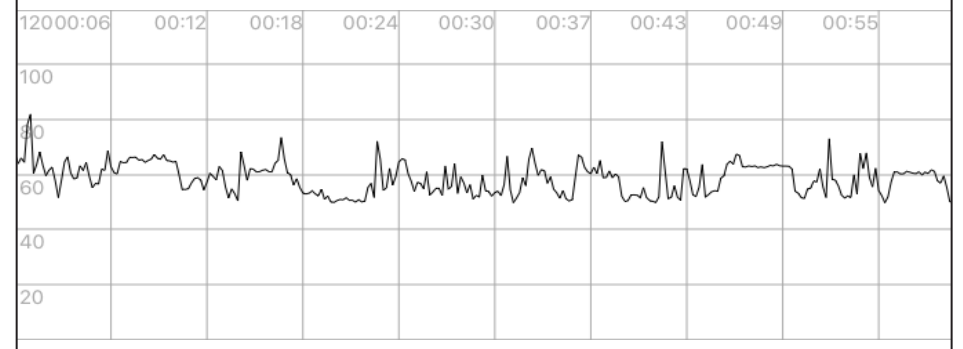
#### Measurement configurations

|                     |             |
|---------------------|-------------|
| Frequency Weighting | A           |
| Response Time       | Fast (0.2s) |
| Calibration         | 0.0 dB      |

#### Measurement results

|          |         |
|----------|---------|
| Duration | 1m:0s   |
| Avg/Leq  | 63.7 dB |
| MIN      | 49.9 dB |
| MAX      | 82.1 dB |
| PEAK     | 91.6 dB |
| TWA      | 0.0 dB  |
| Dose     | 0.0 %   |

#### Graph



## Appendix 3 (Source 1)

### Interview Result 1

Chaewon Lee, Senior at Indiana University

1. Do you listen to music while studying? Sometimes.
2. What type of music do you listen to while studying? White noise or music without lyrics.
3. When you listen to music while studying, does it help you? It helps me focus especially when there are other sounds in the environment like people talking loudly.
4. How loud do you listen to your music while studying? Pretty quiet (on a scale of 1-10 probably 3 or 5).
5. Why do you listen to your music while studying? Having background music can cancel out other noise and can help me focus better than when it is completely silent.
6. What feature do you wish your earphones had? Noise cancelling.
7. Do you have a preset playlist while studying? No.
8. Do you listen to songs that you're familiar with or are you more adventurous with your choice of music? Songs I'm more familiar with.
9. How often do you listen to music? Everyday, especially walking around campus.

### Yes or No Questions:

1. Would a quieter environment help you concentrate? Yes
2. Would you mind if your earphones auto-adjusted the volume for you? Yes
3. When you study do you stay in one place for a long period of time? Yes
4. Would you say music helps you concentrate on your work? No
5. When you're looking for a place to study, do you like to isolate yourself? Yes

## Appendix 4 (Source 2)

### Interview Result 2

Elise Jane Lee, Junior at Indiana University

1. Do you listen to music while studying? yes, I enjoy listening to music while studying. I feel that it helps cancel out background noise which helps me focus on studying. What type of music do you listen to while studying?
2. When you listen to music while studying, does it help you? classical or pop
3. How loud do you listen to your music while studying? When I know I need to sit down and focus for long periods of time during the day, listening to classical music helps me a lot. However, if I have to study at night and I'm feeling tired, pop music can help me stay awake. If I'm listening to pop music it can be distracting since I enjoy the music. Rather than studying, I can find myself focusing on the song rather than my work.
4. How loud do you listen to your music while studying? When I'm studying in a public area, I like to play my music loudly since it can block out other noise. At this point, I'm used to playing my music loudly, so even when I'm at home and it's quiet I'll still listen to music at a high level.
5. Why do you listen to your music while studying? I like to listen to music while studying because it blocks out any surrounding noise
6. What feature do you wish your earphones had? Since I typically study in a public place, I wish my earphones would have a noise cancelling feature.
7. Do you have a preset playlist while studying? Yes, Spotify is my favorite music app because people can create playlists, and that's where I get my study playlist.
8. Do you listen to songs that you're familiar with or are you more adventurous with your choice of music? I like to listen to a variety of music, even if the music isn't in English I can still enjoy the beat and melody.
9. How often do you listen to music? I typically listen to music everyday. Even if I'm not studying, I like to listen to music when I'm walking to class or cleaning my apartment.

### Yes or No Questions:

1. Would a quieter environment help you concentrate? Yes
2. Would you mind if your earphones auto-adjusted the volume for you? Yes
3. When you study do you stay in one place for a long period of time? Yes
4. Would you say music helps you concentrate on your work? Yes
5. When you're looking for a place to study, do you like to isolate yourself? No

## Appendix 5 (Source 3)

### Interview Result 3

Megan Jerrells, Sophomore at Indiana University

1. Do you listen to music while studying? Yes.
2. What type of music do you listen to while studying? Mainly low-fi or downbeat music.
3. When you listen to music while studying, does it help you? Yes. Music helps me cancel out other unnecessary background noises. Having music as a constant steady background noise helps me focus. I usually just listen to one song on repeat, giving my brain is steady rhythm to follow rather than being distracted by other things.
4. How loud do you listen to your music while studying? Not too loud. Loud enough to hear, but it is just background noise.
5. Why do you listen to your music while studying? It helps me focus on what I am doing. I know it seems like it would be a distraction, but actually the noises that aren't the music distract me, so the music cancels everything out.
6. What feature do you wish your earphones had? I would like to own airpods but I am poor :( .
7. Do you have a preset playlist while studying? Yes. I have a preset playlist for studying because it helps me get in the right mindset. This music usually does not include songs with lyrics. I will adventure with new music, but not when I am studying.
8. Do you listen to songs that you're familiar with or are you more adventurous with your choice of music? Mainly the ones I am familiar with.
9. How often do you listen to music? Everyday, multiple times a day.

### Yes or No Questions:

1. Would a quieter environment help you concentrate? Yes
2. Would you mind if your earphones auto-adjusted the volume for you? Yes
3. When you study do you stay in one place for a long period of time? Yes
4. Would you say music helps you concentrate on your work? Yes
5. When you're looking for a place to study, do you like to isolate yourself? No

## Appendix 6 (Source 4)

### Interview Result 4

Heemean Kim, Senior at Indiana University

1. Do you listen to music while studying? Yes
2. What type of music do you listen to while studying? Classical music or Lo-fi hiphop
3. When you listen to music while studying, does it help you? It helps to get into the zone.
4. How loud do you listen to your music while studying? Loud enough to cancel out outside noise
5. Why do you listen to your music while studying? It helps me to focus.
6. What feature do you wish your earphones had? Noise canceling - Auto volume adjustment
7. Do you have a preset playlist while studying? Yes
8. Do you listen to songs that you're familiar with or are you more adventurous with your choice of music? Adventurous
9. How often do you listen to music? Almost all day long

### Yes or No Questions:

1. Would a quieter environment help you concentrate? Yes
2. Would you mind if your earphones auto-adjusted the volume for you? Yes
3. When you study do you stay in one place for a long period of time? Yes
4. Would you say music helps you concentrate on your work? Yes
5. When you're looking for a place to study, do you like to isolate yourself? No